

Organic Chemistry

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Southeast University

School of Chemistry and Chemical Engineering

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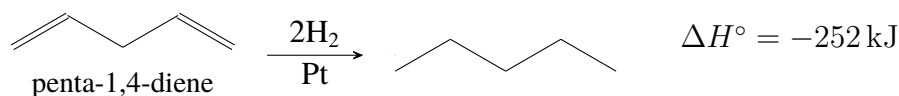
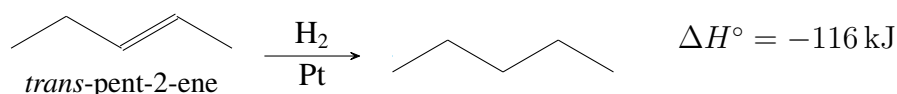
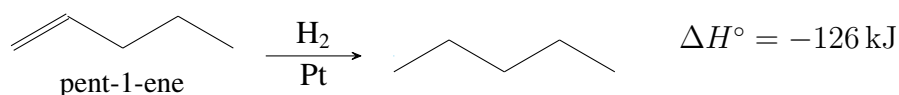
15 Conjugated Systems, Orbital Symmetry, and Ultraviolet Spectroscopy

15.1 Introduction: Conjugated Systems 20260303

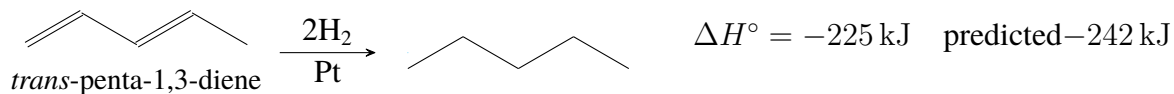
1. Conjugated double bonds: separated by one single bond.
2. Isolated double bonds: separated by two or more single bonds.
3. Cumulated double bonds: allenes, adjacent to each other.
4. Conjugated double bonds are more stable than isolated double bonds.

15.2 Stabilities of Dienes 20260303

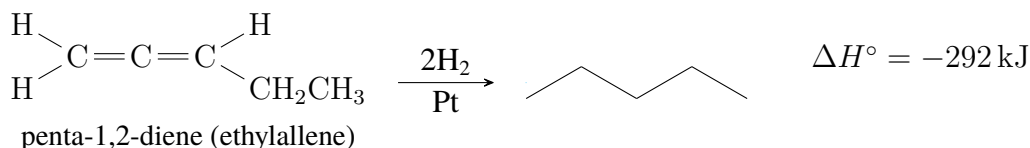
1. Using heats of hydrogenation to compare the relative stabilities of alkenes. The more stable the compound, the less exothermic during hydrogenation.



2. For conjugated dienes, the heat of hydrogenation is less than the sum for the individual double bonds, showing that the conjugated diene has extra stability.



3. The heat of hydrogenation of penta-1,2-diene is larger than any of the other pentadienes.



4. (1) The cumulated double bonds are less stable than isolated double bonds and much less stable than conjugated double bonds.
(2) Corresponding alkynes are less stable than alkenes.
(3) More substitution on double bonds or triple bonds increases the stability.

Example 15.1 Rank each group of compounds in order of increasing heat of hydrogenation.

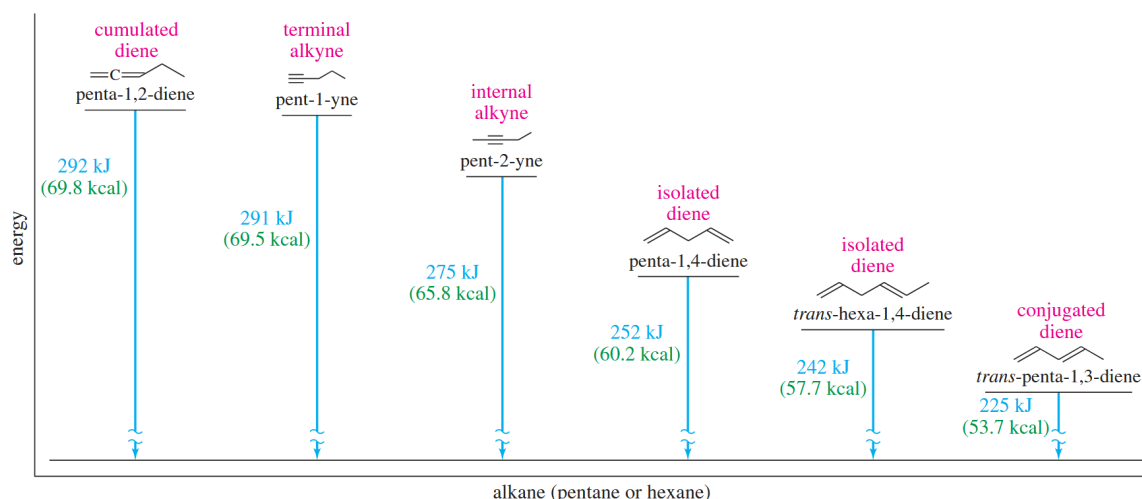


Figure 15.1 Relative Stabilities of Dienes and Alkynes

hexa-1,2-diene; hexa-1,3,5-triene; hexa-1,3-diene; hexa-1,4-diene; hexa-1,5-diene; hexa-2,4-diene

Solution: to be filled with chemfig

Example 15.2 In a strongly acidic solution, cyclohexa-1,4-diene tautomerizes to cyclohexa-1,3-diene. Propose a mechanism for this rearrangement, and explain why it is energetically favorable.

Solution: to be filled with chemfig

15.3 Molecular Orbital Picture of a Conjugated System 20260303 20260305

The extra stability of conjugated systems can be explained by examining their molecular orbitals.

15.3.1 Structure and Bonding of Buta-1,3-diene

1. The 1.48 Å central carbon–carbon single bond is shorter than the 1.54 Å bonds typical of alkanes.
2. Electrons are delocalized over molecules, creating some pi overlap and pi bonding in the bond.

15.3.2 Constructing the Molecular Orbitals of Buta-1,3-diene

1. Molecular orbitals(MO)

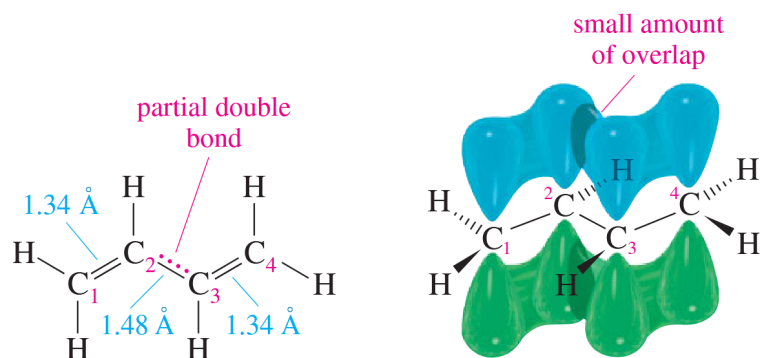


Figure 15.2 Structure of Buta-1,3-diene

2. bonding, nonbonding, and antibonding orbitals
3. energies symmetrically distributed
4. energy of isolated p orbital in the middle of the MO energy levels
5. The bonding MOs are filled and the antibonding MOs are vacant.
6. The average energy of the electrons is slightly lower in butadiene. This lower energy is the resonance stabilization of the conjugated diene.
7. Conformations of 1,3-Butadiene: *s*-trans and *s*-cis, the former, less crowded, is more stable than the latter.
8. The *s*-cis and *s*-trans conformers of butadiene (and all the skew conformations in between) easily interconvert at room temperature (The basis of DA reaction).

15.4 Allylic Cations 20260305

1. Conjugated compounds undergo a variety of reactions, many of which involve intermediates that retain some of the resonance stabilization of the conjugated system.
2. allyl group: $\text{CH}_2=\text{CH}-\text{CH}_2-$
3. Allylic cations and radicals are stabilized by delocalization.
4. Double bond interact with cations(exchange).
5. Stability of carbocations: $\text{H}_3\text{C}^+ < 1^\circ < 2^\circ$, allyl $< 3^\circ$, substituted allylic

15.5 1,2- and 1,4-Addition to Conjugated Dienes 20260305

1. Electrophilic addition to the double bond produces the most stable intermediate.
2. For conjugated dienes, the intermediate carbocation is a resonance stabilized allylic cation.
3. MECHANISM 15-1 1,2- and 1,4-Addition to a Conjugated Diene
 - (1) Protonation of one of the double bonds forms a resonance-stabilized allylic cation.

- (2) A nucleophile attacks at either electrophilic carbon atom.

15.6 Kinetic versus Thermodynamic Control in the Addition of HBr to Buta-1,3-diene 20260305

1. $-80\text{ }^{\circ}\text{C}$ 1,2-addition (kinetic control), $40\text{ }^{\circ}\text{C}$ 1,4-addition (thermodynamic control)
2. The 1,2-addition will be the faster addition at any temperature (lower E_a , secondary carbocation intermediate), but the 1,4-addition will be more stable (higher E_a , but more stable, more substituted alkene product).
3. The first step of the reaction is kinetically controlled, the second step is thermodynamically controlled.
4. Thermodynamic control at $40\text{ }^{\circ}\text{C}$
5. Kinetic control at $-80\text{ }^{\circ}\text{C}$

Example 15.3 hint: In many cases, an allylic carbocation is more stable than a bromonium ion.

15.7 Allylic Radicals 20260305

1. Allylic radicals are stabilized by resonance delocalization.
2. In a free-radical bromination of cyclohexene, substitution occurs entirely at the allylic position.
3. MECHANISM 15-2 Free-Radical Allylic Bromination
 - (1) Initiation: Formation of radicals.

Note Cleavage should happen in nonpolar solvents. e.g. CCl_4

4. Stability of free radicals: $\text{CH}_3 < 1^{\circ} < 2^{\circ} < 3^{\circ} < \text{allylic} < \text{benzylic}$
5. Bromination Using N-Bromosuccinimide (NBS)

Example 15.4 prob 15-11

15.8 Molecular Orbitals of the Allylic System 20260305

15.9 Electronic Configurations of the Allyl Radical, Cation, and Anion 20260305

15.10 SN2 Displacement Reactions of Allylic Halides and Tosylates 20260305

1. Allyl bromide reacts with nucleophiles by the mechanism about 40 times faster than n-propyl bromide.
2. Allylic halides are so reactive that they couple with Grignard and organolithium reagents

15.11 The Diels–Alder Reaction 20260310

1. Otto Diels and Kurt Alder, alkenes and alkynes with electron-withdrawing groups (electron-deficient alkene, dienophile) add to conjugated dienes to form sixmembered rings (cyclohexene)

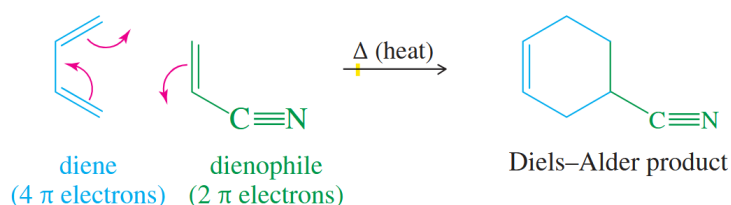


Figure 15.3 The Diels–Alder Reaction

2. a $[4 + 2]$ cycloaddition

Mechanism 15.1

15.11.1 Stereochemical Requirements of the Diels–Alder Transition State

1. *s*-cis Conformation of the Diene
 - (1) The diene must be in the *s*-cis conformation to react.
 - (2) Dienes with functional groups that hinder the *s*-cis conformation react more slowly than butadiene. Dienes with functional groups that hinder the *s*-trans conformation react faster than butadiene.
2. syn Stereochemistry
 - (1) There is no opportunity for any of the substituents to change their stereochemical positions during the course of the reaction.
3. The Endo Rule
 - (1) Dienophile has a pi bond in its electron-withdrawing group (as in a carbonyl group or a cyano group).
 - (2) Secondary overlap: an overlap of the *p* orbitals of the electron-withdrawing group with the *p* orbitals of C2 and C3 of the diene.

- (3) Secondary overlap helps to stabilize the transition state.
- (4) This position is called the endo position, and This stereochemical preference for the electron-withdrawing substituent to appear in the endo position is called the endo rule.

15.11.2 Diels–Alder Reactions Using Unsymmetrical Reagents

1. When the diene and dienophile are both unsymmetrically substituted, the DA reaction usually gives a single product (or a major product).
2. Predict the major product by considering how the substituents polarize the diene and the dienophile in their charge-separated resonance forms.
3. An electron-donating substituent (D) on the diene and an electron-withdrawing substituent (W) on the dienophile usually show either a 1,2- or 1,4-relationship in the product.

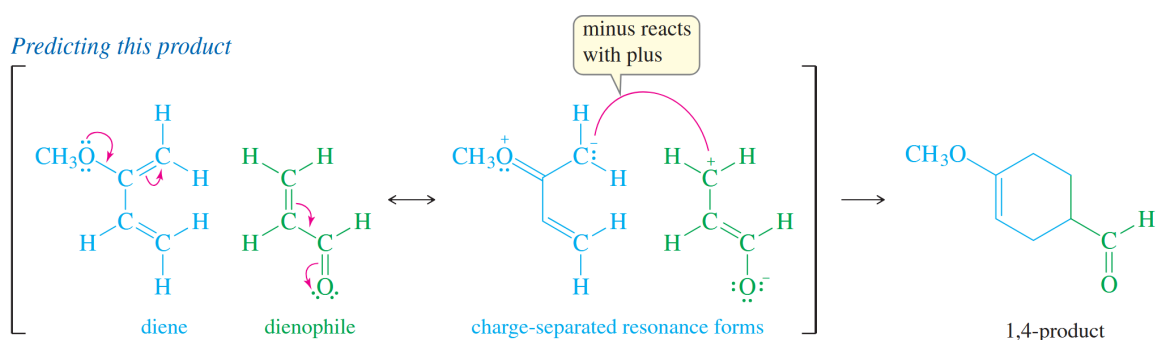


Figure 15.4 Formation of 1,4-product

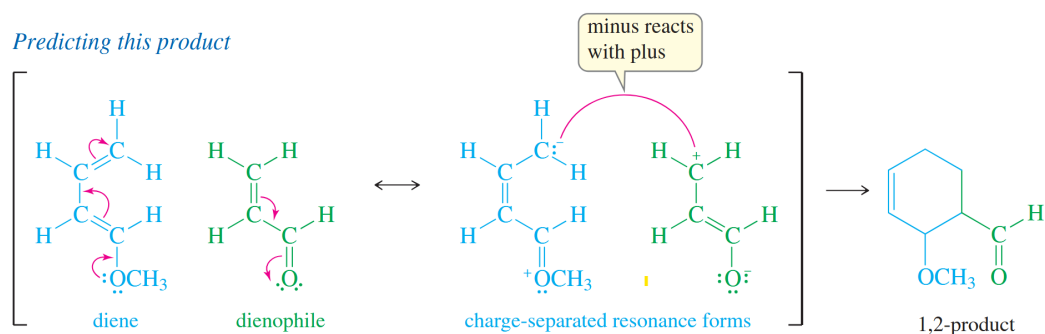


Figure 15.5 Formation of 1,2-product

4. Remember the major product will not be a 1,3-product.

15.12 The Diels–Alder as an Example of a Pericyclic Reaction 20260310 20260312

1. Pericyclic reactions involve the concerted forming and breaking of bonds within a closed ring of interacting orbitals.
2. Conservation of orbital symmetry: the molecular orbitals of the reactants must flow smoothly into the MOs of the products without any drastic changes in symmetry.

15.12.1 Conservation of Orbital Symmetry in the Diels–Alder Reaction

1. Highest Occupied Molecular Orbital (HOMO), Lowest Unoccupied Molecular Orbital (LUMO)
2. The HOMO of butadiene has the correct symmetry (plus with plus and minus with minus) to overlap in phase with the LUMO of ethylene.
3. The bonding interactions stabilize the transition state and promote the concerted reaction. This favorable result predicts that the reaction is symmetry-allowed.

15.12.2 The “Forbidden” [2+2] Cycloaddition

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15.12.3 Photochemical Induction of Cycloadditions

1. This higher orbital, formerly the LUMO, is now occupied: It is the new the HOMO*, the HOMO of the excited molecule.

Example 15.5 Show that the [4 + 2] Diels–Alder reaction is photochemically forbidden. (In basic chemistry not react, but may react in different symmetry gouxiang)

15.13 Ultraviolet Absorption Spectroscopy 20260312

1. Ultraviolet (UV) spectroscopy detects the electronic transitions of **conjugated systems** and provides information about the length and structure of the conjugated part of a molecule
2. UV spectroscopy giving more specialized information than does IR or NMR.

15.13.1 Spectral Region

1. jadsfkjas

15.13.2 Ultraviolet Light and Electronic Transitions

1. A photon with the right amount of energy can excite an electron from the bonding orbital (π) to the antibonding orbital (π^*). This transition from a bonding orbital to a antibonding orbital is called a $\pi \rightarrow \pi^*$ transition.
2. In conjugated systems, there are electronic transitions with lower energies that correspond to wavelengths longer than 200 nm.
3. In general, the energy difference between HOMO and LUMO decreases as the length of conjugation increases.

15.13.3 Obtaining an Ultraviolet Spectrum

1. The sample solution is placed in a quartz cell, and some of the solvent is placed in a reference cell.
2. Compare the amount of light transmitted through the sample (the sample beam) with the amount of light in the reference beam. Measure the intensity ratio of the reference (I_r) beam compared with the sample beam (I_s).
3. The absorbance, A , of the sample at a particular wavelength is governed by *Beer's law*.

$$A = \log \frac{I_r}{I_s} = \epsilon cl$$

4. Molar absorptivity (ϵ) is a measure of how strongly the sample absorbs light at that wavelength.
5. The molar absorptivity of onjugated system is about 5000 to 50000.

15.13.4 Interpreting UV-Visible Spectra

1. buta-1,3-diene 217 nm, cyclohexa-1,3-diene 256 nm.
2. The Woodward-Fieser Rules
3. the values of increase by about 30 nm to 40 nm for each double bond extending the conjugated system. Alkyl groups also increase the value of by about 5 nm per alkyl group.

Example 15.6

end of Ch 15

16 Aromatic Compounds

16.1 Introduction: The Discovery of Benzene 20260312 20260317

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16.2 The Structure and Properties of Benzene 20260312 20260317

1. Kekulé structure, failed to explain existence of only one isomer of 1,2-dichlorobenzene.
2. The Resonance Representation: a resonance hybrid between the two Kekulé structures, represented by drawing a circle inside the six-membered ring as a combined representation.
3. Structure of Benzene: Each sp^2 hybridized C in the ring has an unhybridized p orbital perpendicular to the ring which overlaps around the ring.
4. An aromatic compound is a cyclic compound containing some number of conjugated double bonds and having an unusually large resonance energy.
5. The Unusual Reactions of Benzene: will not undergo the typical reactions of polyenes.
6. **Unusual Addition of Bromine to Benzene**
 - (1) When bromine adds to benzene, a catalyst such as **FeBr₃** is needed.
 - (2) Substitution of a hydrogen by bromine.
 - (3) No addition of Br₂ to the double bond.
7. The difference between the predicted and the observed value of hydrogenation is called the resonance energy.
8. Annulenes
 - (1) Annulenes are hydrocarbons with alternating single and double bonds.
 - (2) Cyclobutadiene is [4]annulene, Benzene is [6]annulene Cyclooctatetraene is [8]annulene.
 - (3) Reactivity: Cyclobutadiene is so reactive that it dimerizes before it can be isolated. Cyclooctatetraene adds Br₂ readily to the double bonds.
 - (4) tub conformation of cyclooctatetraene

16.3 The Molecular Orbitals of Benzene 20260317

1. Six overlapping p orbitals must form six molecular orbitals.
Three will be bonding, three antibonding.
Lowest energy MO will have all bonding interactions, no nodes.
As energy of MO increases, the number of nodes increases.

2. First MO of Benzene

Intermediate MO of Benzene

3. **Electronic Energy Diagram** for Benzene: Six electrons fill three bonding pi orbitals. All bonding orbitals are filled (“closed shell”), an extremely stable arrangement.

16.4 The Molecular Orbital Picture of Cyclobutadiene 20260317

1. Electronic Energy Diagram for Cyclobutadiene: Following Hund’s rule, two electrons are in separate nonbonding molecular orbitals. This diradical would be very reactive.
2. The Polygon Rule: The molecular orbital energy diagram of a regular, completely conjugated cyclic system has the same polygonal shape as the compound, with one vertex (the all-bonding MO) at the bottom.
3. However, cyclooctatetraene will not follow the Polygon Rule (planar) but will form a tub conformation to be more stable.

16.5 Aromatic, Antiaromatic, and Nonaromatic Compounds 20260317

1. Aromatic compounds are those that meet the following criteria:
 - (1) Structure must be cyclic with conjugated pi bonds.
 - (2) Each atom in the ring must have an unhybridized p orbital (sp^2 or sp).
 - (3) The p orbitals must overlap continuously **around the ring**. Structure must be planar (or close to planar for effective overlap to occur).
 - (4) Delocalization of the pi electrons over the ring must lower the electronic energy.
2. An antiaromatic compound is one that meets the first three criteria, but delocalization of the pi electrons over the ring increases the electronic energy. Compounds or ions.
3. Nonaromatic compounds do not have a continuous ring of overlapping p orbitals and may be nonplanar.

16.6 Hückel’s Rule 20260317

1. To qualify as aromatic or antiaromatic, a cyclic compound must have a continuous ring of overlapping p orbitals, usually in a planar conformation.
2. Once the criteria are met, Hückel’s rule applies:

If the number of pi electrons in the cyclic system is: the system is aromatic. ($4N$) the system is antiaromatic. N is an integer, commonly 0, 1, 2, or 3.
3. $4N$ Annulenes

- (1) [4]annulene is antiaromatic.
 - (2) [8]annulene would be antiaromatic, but it's not planar, so it's nonaromatic.
 - (3) Larger $4N$ annulenes are not antiaromatic because they are flexible enough to become nonplanar.
4. $4N+2$ Annulenes
- (1) [6]annulene (benzene) is aromatic.
 - (2) [10]annulene is aromatic except for the isomers that are nonplanar. (two hydrogen atoms interfere with each other)
 - (3) Some larger $4N+2$ annulenes can achieve planar conformation, they are aromatic.

16.7 MO Derivation of Hückel's Rule 20260317

1. Aromatic compounds have $(4N+2)$ electrons and the orbitals are filled.
2. Antiaromatic compounds have only $4N$ electrons and has unpaired electrons in two degenerate orbitals.

16.8 Aromatic ions 20260317 20260319

16.8.1 Cyclopentadienyl Ions

1. The cation has an empty p orbital, 4 electrons, so it is antiaromatic.
2. The anion has a nonbonding pair of electrons in a p orbital, 6 electrons, it is aromatic.
3. Cyclopentadiene is acidic because deprotonation will convert it to an aromatic ion.
4. Deprotonation of Cyclopentadiene:
By deprotonating the sp^3 carbon of cyclopentadiene, the electrons in the p orbitals can be delocalized over all five carbon atoms and the compound would be aromatic.
Cyclopentadiene is acidic because deprotonation will convert it to an aromatic ion.
5. Resonance Forms of Cyclopentadienyl Ions
The resonance picture gives a **misleading suggestion** of stability.

16.8.2 The Cycloheptatrienyl Ions

1. The cycloheptatrienyl cation has 6 π electrons and an empty p orbital.
2. The cycloheptatrienyl cation is easily formed by treating the corresponding alcohol with dilute (0.01 molar) aqueous sulfuric acid.
3. The cycloheptatrienyl cation is called the tropylium ion. This aromatic ion is much less reactive than most carbocations.

16.8.3 The Cyclooctatetraene Dianion

1. Cyclooctatetraene reacts with potassium metal to form an aromatic dianion.
2. The dianion has 10 pi electrons and is aromatic.
3. Biphenyl has the following structure. p752
4. Hexahelicene p754

Example 16.1 Explain why each compound or ion should be aromatic, antiaromatic, or nonaromatic.

1. the cyclononatetraene cation, anti
2. the cyclononatetraene anion, aro
3. the [16]annulene dianion, aro if planar
4. the [18]annulene dianion, anti
5. non
6. the [20]annulene dication, aro if planar

Example 16.2 The following hydrocarbon has an unusually large dipole moment. Explain how a large dipole moment might arise.

SOLUTION The reason for the dipole can be seen in a resonance form distributing the electrons to give each ring 6 pi electrons. This resonance picture gives one ring a negative charge and the other ring a positive charge.

Example 16.3 When 3-chlorocyclopropene is treated with AgBF_4 , AgCl precipitates. The organic product can be obtained as a crystalline material, soluble in polar solvents such as nitromethane but insoluble in hexane. When the crystalline material is dissolved in nitromethane containing KCl , the original 3-chlorocyclopropene is regenerated. Determine the structure of the crystalline material, and write equations for its formation and its reaction with chloride ion.

Example 16.4 The polarization of a carbonyl group can be represented by a pair of resonance structures:

16.8.4 Summary of Annulenes and Their Ions

1. The 2, 6, and 10 electron systems are aromatic, while the 4 and 8 electron systems are antiaromatic if they are planar.

16.9 Heterocyclic Aromatic Compounds 20260319

16.9.1 Pyridine

1. Pyridine has a nitrogen atom in place of one of the six C—H units of benzene.
2. Pyridine has a six-membered heterocyclic ring with six pi electrons.
3. The nonbonding electrons are in an sp^2 hybrid orbital in the plane of the ring, perpendicular to the pi system and do not overlap with it.
4. Pyridine is basic because of its available pair of nonbonding electrons.
5. The pyridinium ion is still aromatic because the additional proton has no effect on the electrons of the aromatic sextet.

16.9.2 Pyrrole

1. Pyrrole is an aromatic five-membered heterocycle, with one nitrogen atom and two double bonds.
2. The pyrrole nitrogen atom is sp^2 hybridized, and has a lone pair of electrons in the unhybridized p orbital.
3. The unhybridized orbital overlaps with the p orbitals of the carbon atoms to form a continuous ring.
4. Pyrrole is a much weaker base than pyridine. To form a bond to a proton requires the use of one of the electron pairs in the aromatic sextet.
5. Basicity: Delocalized lone pair of electrons is not basic, usually with a hydrogen atom on the nitrogen atom.

16.9.3 Pyrimidine and Imidazole

1. Pyrimidine is a six-membered heterocycle with two nitrogen atoms situated in a 1,3- arrangement.
2. Imidazole is an aromatic five-membered heterocycle with two nitrogen atoms.
3. Purine has an imidazole ring fused to a pyrimidine ring. Purine has three basic nitrogen atoms and one pyrrole-like nitrogen.

Example 16.5 The proton NMR spectrum of 2-pyridone gives the chemical shifts shown.

16.9.4 Furan and Thiophene 20260319

1. Furan is an aromatic five-membered heterocycle, and the heteroatom is oxygen.
2. Thiophene is similar to furan, with a sulfur atom in place of the furan oxygen. The sulfur atom uses an unhybridized $3p$ orbital to overlap with the $2p$ orbitals on the carbon atoms.

Example 16.6 Explain why each compound is aromatic, antiaromatic, or nonaromatic.

Example 16.7 Borazole, $B_3N_3H_6$, is an unusually stable cyclic compound. Propose a structure for borazole, and explain why it is aromatic.

16.10 Polynuclear Aromatic Hydrocarbons 20260324

1. Composed of two or more fused benzene rings, sharing two carbon atoms and the bond between them.
2. Naphthalene: the simplest fused aromatic compound, consisting of two fused benzene rings.
3. Anthracene and Phenanthrene: As the number of fused aromatic rings increases, the resonance energy per ring continues to decrease and the compounds become more reactive.
4. Larger Polynuclear Aromatic Hydrocarbons: have more fused rings than anthracene and phenanthrene, and they have less resonance energy per ring and are more reactive.

16.11 Aromatic Allotropes of Carbon 20260324

16.11.1 Allotropes of Carbon: Diamond

1. Amorphous carbon refers to charcoal, soot, coal, and carbon black.
2. Diamond has a crystalline structure containing tetrahedral carbon atoms linked together in a three-dimensional lattice.

16.11.2 Graphite

1. Graphite has the layered planar structure.
2. Each layer of graphite is a nearly infinite lattice of fused aromatic rings. No bonds but van der Waals forces hold the layers together.

16.11.3 Fullerenes

16.12 Fused Heterocyclic Compounds 20260324

Electrostatic potential map shows electron-rich (red) and electron-poor (blue).

16.13 Nomenclature of Benzene Derivatives 20260324

1. The following compounds are usually called by their historical common names, and almost never by the systematic IUPAC names:
phenol (benzenol), toluene (methylbenzene), aniline (benzenamine), anisole (methoxybenzene), styrene (vinylbenzene), acetophenone (methyl phenyl ketone), benzaldehyde, benzoic acid
2. Many compounds are named as derivatives of benzene.
tert-butylbenzene, nitrobenzene, ethynylbenzene (phenylacetylene), benzenesulfonic acid
3. Disubstituted benzenes are named using the prefixes **ortho-** (1,2), **meta-** (1,3), and **para-** (1,4) to specify the substitution patterns. These terms are abbreviated *o*-, *m*-, and *p*-.
4. With three or more substituents on the benzene ring, numbers are used to indicate their positions, and give the lowest possible numbers to the substituents. The carbon atom bearing the functional group that defines the base name.
5. Many disubstituted benzenes (and polysubstituted benzenes) have historical names.
m-xylene,
6. When the benzene ring is named as a substituent on another molecule, it is called a **phenyl group**.

16.14 Physical Properties of Benzene and Its Derivatives 20260324

16.15 Spectroscopy of Aromatic Compounds 20260324 20260326

16.15.1 Infrared Spectroscopy

1. Aromatic compounds are readily identified by their infrared spectra because they show a characteristic C=C stretch around 1600 cm^{-1} .
2. s: strong, b: broad, m: medium
3. 4000 - 2500 (X—H, O—H, N—H, C—H),
2500 - 2000 (C≡C, C≡N),
2000 - 1500 (C=C, C=O),
1500 - 400 (fingerprint)

16.15.2 NMR Spectroscopy

16.15.3 Mass Spectrometry

1. The benzyl cation may rearrange to give the aromatic tropylium ion.
2. Alkylbenzenes frequently give ions corresponding to the tropylium ion at m/z 91.

16.15.4 Ultraviolet Spectroscopy

1. The most characteristic part of the spectrum is the band centered at 254 nm, called the **benzenoid band**.
2. About three to six small, sharp peaks (called fine structure) usually appear in this band.

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17 Reactions of Aromatic Compounds

17.1 Electrophilic Aromatic Substitution 20260326

1. Although benzene's pi electrons are in a stable aromatic system, they are available to attack a strong electrophile to give a carbocation.
2. This resonance-stabilized carbocation is called a **sigma complex** because the electrophile is joined to the benzene ring by a new sigma bond.
3. The sigma complex (also called an arenium ion) is not aromatic because the sp^3 hybrid carbon atom interrupts the ring of p orbitals.

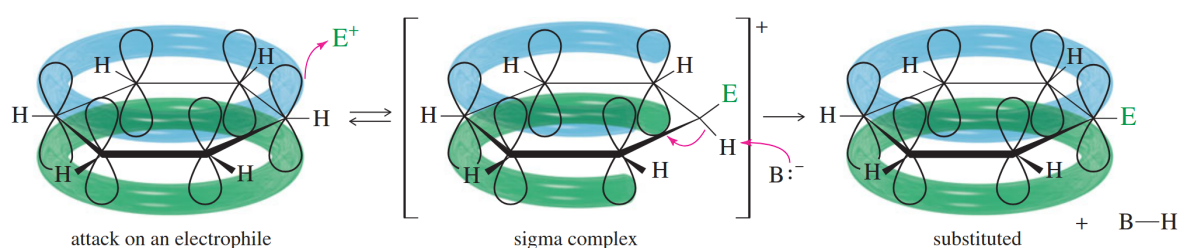


Figure 17.1 Sigma Complex

Mechanism 17.1 Electrophilic Aromatic Substitution

- Step 1: Attack on the electrophile forms the sigma complex.
- Step 2: Loss of a proton regains aromaticity and gives the substitution product.

Example 17.1 Iodination of toluene

17.2 Halogenation of Benzene 20260326

17.2.1 Bromination of Benzene

1. Bromination follows the general mechanism for electrophilic aromatic substitution.
2. Bromine itself is not sufficiently electrophilic, and the formation of Br^+ is difficult.
3. A strong Lewis acid such as FeBr_3 catalyzes the reaction by forming a complex with Br_2 that reacts like Br^+ .

Mechanism 17.2 Bromination of Benzene

17.2.2 Chlorination of Benzene

1. Chlorination of benzene works much like bromination, except that aluminum chloride (AlCl_3) is most often used as the Lewis acid catalyst.

17.2.3 Iodination of Benzene

1. Iodination of benzene requires an acidic oxidizing agent, such as nitric acid, to produce iodide cation.

Example 17.2 fluorination

17.3 Nitration of Benzene 20260326

1. Benzene reacts with hot, concentrated nitric acid to give nitrobenzene. A safer and more convenient procedure uses a mixture of nitric acid and sulfuric acid.
2. Sulfuric acid is a catalyst, allowing nitration to take place more rapidly and at lower temperatures.

Mechanism 17.3 Nitration of Benzene

- Preliminary steps: Formation of the nitronium ion, NO_2^+ .
 - Step 1: Attack on the electrophile forms the sigma complex.
 - Step 2: Loss of a proton gives nitrobenzene.
3. Aromatic nitro groups are easily reduced to amino (—NH_2) groups by treatment with an active metal such as tin, zinc, or iron in dilute acid.
 4. Nitration followed by reduction is often the best method for adding an amino group to an aromatic ring.

17.4 Sulfonation of Benzene 20260326

17.4.1 Sulfonation

1. Sulfur trioxide (SO_3) is the electrophile.
2. “Fuming sulfuric acid” is the common name for a solution of 7% SO_3 in H_2SO_4 .

Mechanism 17.4 Sulfonation of Benzene

17.4.2 Desulfonation

1. A sulfonic acid group may be removed from an aromatic ring by heating in dilute sulfuric acid.
2. Steam is often used as a source of both water and heat for desulfonation.

17.4.3 Protonation of the Aromatic Ring; Hydrogen–Deuterium Exchange

17.5 Nitration of Toluene: The Effect of Alkyl Substitution 20260326

1. Toluene (methylbenzene) reacts with a mixture of nitric and sulfuric acids much like benzene does, but with some interesting differences:
 - (1) Toluene reacts about 25 times faster than benzene under the same conditions. Toluene is activated toward electrophilic aromatic substitution, and the methyl group is an **activating group**.
 - (2) Nitration of toluene gives a mixture of products, primarily those resulting from substitution at the ortho and para positions. The methyl group of toluene is an **ortho, para-director**.
2. In ortho or para substitution of toluene, the positive charge is spread over two secondary carbons and one tertiary (3°) carbon (bearing the CH_3 group).
3. The sigma complex for meta substitution has its positive charge spread over three 2° carbons.